

MVP: the clever covariate on the next step

2026

One beat is one committed step. $Y = 1$ when that episode’s search returns a solved state, and 0 otherwise. Y is written onto the committed steps only after the episode returns. The query beat has X and the action prior e . It does not have Y . After the write-back,

$$H = \frac{Y - e}{e(1 - e)},$$

with e clipped to $(0.001, 0.999)$. The online anomaly score is $|H|$. $H > 0.5$ is aggressive, $H < -0.5$ is conservative, and the rest is keep. No UCB term is added.

On a binary Y , a success gives $H = 1/e > 0$ and a failure gives $H = -1/(1 - e) < 0$. The sign is the detector. When $e < 1/2$, a success has the larger absolute value, so $|H|$ alone is not a failure score. The step that produced Y cannot read its own H . The next episode can.

The read is tested on the three-offer search. Forty users, drift at user 16, three children each. The habit offer activates half of the users. After the drift the judge selects click and the activation rate falls from 0.50 to 0.167. The next-user rule stays on habit through the first 12 users. It switches when the mean loss of the last four finished users exceeds the reference mean by 0.15. That is the same information as a run of conservative H , because a failure is exactly $H < 0$. The switch occurs at user 17.

Policy	Pre-drift Y	Post-drift Y	Reading
Judge, H not read	0.500	0.167	shortfall, -8 users
Next user reads the sign	0.500	0.625	increment, $+3$ users

The second line activates 11 more of the 24 post-drift users than the drifted judge, and 3 more than the pre-drift rate. Children stay at 3. The prior on the selected offer moves from 0.50 to 0.65. The conservative share moves from 0.50 to 0.375. $|H|$ stays near 2. Writing H into the prompt, without letting the next argmax read the sign, does not move Y . The increment is the read.